

Shock-Turbulence Interaction in Variable Density Flows



Yifeng Tian, Farhad Jaberri and Daniel Livescu

Abstract Accurate numerical simulations of shock-turbulence interaction (STI) are conducted by a hybrid monotonicity preserving-compact finite difference scheme for a detailed study of STI in variable density flows. Numerical accuracy of the simulations has been established using a series of grid, particle, and linear interaction approximation (LIA) convergence tests. The results show that for current parameter ranges, turbulence amplification by the normal shock wave is much higher and the reduction in turbulence length scales is more significant when strong density variations exist in STI. The turbulence structure is strongly modified by the shock wave, with a differential distribution of turbulent statistics in regions with different densities. The correlation between rotation and strain is weaker in the multi-fluid case, which is shown to be the result of complex role density plays when the flow passes through the shock wave. Furthermore, a stronger symmetrization of the joint probability density function (PDF) of second and third invariants of the anisotropic velocity gradient tensor (VGT) is observed in the multi-fluid case. Lagrangian dynamics of the VGT and its invariants are studied and the pressure Hessian contributions are shown to be strongly affected by the shock wave and local density, making them important to the flow dynamics and turbulence structure.

Keywords Turbulence · Variable-density flows · Shock waves · Velocity gradient tensor

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1 Introduction

The interaction of a normal shock wave with multi-fluid isotropic turbulence is an extension of the canonical shock-turbulence interaction (STI) problem which includes strong variable density effects. This extended configuration can enhance our understanding of more complex flow problems such as fuel-air mixing in supersonic combustion, the interaction of supernova remnants with interstellar clouds, shock propagation through foams and bubbly liquids, inertial confinement fusion (ICF), and re-shock problem in Richtmyer-Meshkov instability (RMI). Most of the previous theoretical, numerical, and experimental studies of STI have been dedicated to the original canonical problem.

The early theoretical study in Ref. [1] has restricted the STI to the linear interaction regime with a large scale separation between the shock and turbulence, so that the nonlinear and viscous effects are assumed to be negligible during the interaction. By decomposing the pre-shock turbulence into independent modes (acoustic, vortical and entropy) using Kovaznay decomposition [2], the post-shock turbulence statistics can be theoretically derived from the linearized Rankine-Hugoniot jump conditions. This approach is referred to as the linear interaction approximation (LIA) and represents an important limiting case, since it provides analytical predictions for the jumps of fluctuating quantities through the shock.

Due to the challenges of accurate experimental measurements of the smallest time and length scales around the shock wave, numerical simulations have been more widely employed to investigate this interaction. Researchers have been using both shock-capturing and shock-resolving simulations to understand the post-shock amplification of Reynolds stress, vorticity variance, and turbulent length scales [3–10]. Earlier numerical studies have shown limited agreement with the LIA predictions because the parameter range was outside the linear regime. More recently, Ref. [11] have considered a wide range of parameters in their shock-resolving direct numerical simulations (DNS) to show that the DNS results converge to the LIA solutions when the ratio of the shock thickness (δ) to the pre-shock Kolmogorov length scale (η) becomes small. Replacing the actual shock interaction with the LIA relations can extend the reach of DNS to arbitrarily high shock Mach numbers and much larger Taylor Reynolds number (Re_λ) than otherwise computationally feasible, provided that the interaction parameters correspond to the linear regime. This method (named Shock-LIA by the authors) was used for detailed studies of the post-shock turbulent energy flux and vorticity dynamics [12, 13]. Reference [14] used shock-capturing simulation and LIA to study the thermodynamic field generated by STI. In a recent study, Ref. [15] showed, using shock-capturing turbulence-resolving simulations, that the LIA predictions for the Reynolds stresses can be approached (albeit at larger Re_λ values than in full DNS calculations) providing that the scale separation between numerical shock thickness (δ_n) and Kolmogorov length scale is large enough. Thus, when the scale separation is large enough, so that the numerical artifacts near the shock do not influence the flow, the shock-capturing method can properly simulate the STI.

As mentioned above, in many practical applications, STI may occur in a mixture of very different density fluids. This motivated our extension of the canonical STI problem to include variable density effects [15, 16] by considering the pre-shock turbulence as an isotropic mixture of two fluids with different molecular weights, as encountered in non-premixed combustion. Using turbulence-resolving shock-capturing simulations, we have examined the turbulence statistics, turbulence budgets, conditional statistics, and energy spectrum in the multi-fluid STI and found that the nonlinear effects from the density variations significantly change the turbulence properties in both physical and spectral spaces. In another study, Refs. [17, 18] considered a reactive shock wave in a premixed mixture and used LIA and shock-capturing simulations to study the detonation-turbulence interaction. However, there still exist many gaps in our knowledge of the variable density effects on the post-shock turbulence structure and flow topology.

The properties of the velocity gradient tensor (VGT) determine a wide variety of turbulence characteristics, such as the flow topology, deformation of material volume, energy cascade, and intermittency. Understanding both the VGT field immediately after the shock-wave and its dynamics as the flow evolves away from the shock wave are also crucial to the development of subgrid-scale models that can accurately describe the shock interaction and return-to-isotropy effects. Reference [19] has proposed an approach to classify the local flow topology and structure using the invariants of VGT. The dynamical behavior of the VGT has been studied for incompressible flows using the Lagrangian evolution of the invariants along conditional mean trajectories (CMT) [20]. The statistics regarding the invariants of VGT and their Lagrangian dynamics have been used to understand the structure of turbulence in many canonical flows, such as isotropic turbulence, turbulent boundary layer and mixing layers [21–23]. Previous studies on single-fluid STI have examined some of the statistics of the PDF of VGT. References [11, 12] took a step further to investigate the turbulence structure and vorticity dynamics based on the examination of VGT invariants. By taking advantage of the Shock-LIA procedure, they extracted the statistics of VGT and its invariants for a wide range of shock Mach numbers. On the other hand, the dynamics of VGT as the turbulence evolves away from the shock wave cannot be examined using the Shock-LIA procedure. Our preliminary numerical studies on variable density STI have revealed some important features of VGT [24, 25]. The relation between velocity and scalar field has also been studied by Ref. [26] to better understand mixing in this flow configuration. However, these studies have not yet fully revealed the variable density effects on the post-shock turbulence/scalar structure.

This chapter presents a comprehensive study of multi-fluid STI using the turbulence-resolving shock-capturing simulations. The chapter is organized as follows. Details of the simulations are described in Sect. 2 and the testing conducted to assess the accuracy of results are discussed in Sect. 3. Results are presented in Sect. 4.

2 Governing Equations and Solution Procedure

In this section, we will briefly discuss the governing equations and numerical approach used for shock-capturing turbulence-resolving simulations, from which we have extracted the Eulerian and Lagrangian statistics.

2.1 Eulerian Method

The conservative form of the dimensionless compressible Navier-Stokes equations for flows with two miscible species (i.e. continuity, momentum, energy, and species mass fraction transport equations) were solved numerically together with the perfect gas law using a high-order hybrid numerical method [15]. The inviscid fluxes for the transport equations were computed using the fifth-order monotonicity-preserving (MP) scheme, as described in Refs. [15, 27, 28]. The molecular transport terms were calculated using the sixth-order compact scheme [29]. The 3rd-order Runge-Kutta scheme was used for time advancement.

The physical domain for the simulations considered in this paper is a box that has a dimension of 4π in the streamwise direction (denoted as x) and $(2\pi, 2\pi)$ in the transverse directions (denoted as y and z), as shown in Fig. 1a. A buffer layer is used at the end of the computational domain from 4π to 6π to eliminate reflecting waves. In the transverse directions, periodic boundary conditions are used as the flow is assumed to be periodic and homogeneous in these directions. To provide inflow turbulence, pre-generated decaying isotropic turbulence is superposed on the uniform mean flow with Mach number = 2.0 and convected into the domain using Taylor's hypothesis. The inflow turbulent Mach number, Reynolds number and peak wavenumber are $M_t \approx 0.1$, $Re_\lambda \approx 45$, and $k_0 = 4$, respectively. For this M_t value, Taylor's hypothesis is appropriate for approximating spatially developing turbulence with temporally developing turbulence [30]. The variable density (multi-fluid) effects arise from compositional variations, by correlating the density to an isotropic scalar field. The scalar field is generated as a random field following a Gaussian spectrum with a peak at $k_s = 8.0$ and has double-delta probability density function distribution so that the scalar value initially is either 1.0 or 0.0. The initial scalar field is smoothed by solving a diffusion equation so that the scalar field can be fully resolved by the chosen mesh. The resulting scalar field is then allowed to decay in the fully developed isotropic turbulence setup for one eddy turn over time as a passive scalar. The mass fractions of the two fluids can be extracted from the density field using the infinite speed of sound relation [31, 32], which is approximately satisfied at low turbulent Mach numbers. The Atwood number, $A_t = (W_2 - W_1)/(W_2 + W_1)$, calculated from the molar weights of the two fluids, W_1 and W_2 , is 0.28. This value of the Atwood number was chosen such that the variable density effects are non-negligible, yet the interaction with the shock wave is still in the wrinkled-shock regime. The Prandtl number, Pr , and Schmidt number, Sc , are the same and equal to 0.75. The mean flow Mach number, M_s , is 2.0. Immediately before the shock wave, M_t and Re_λ are around 0.09 and 42, respectively.

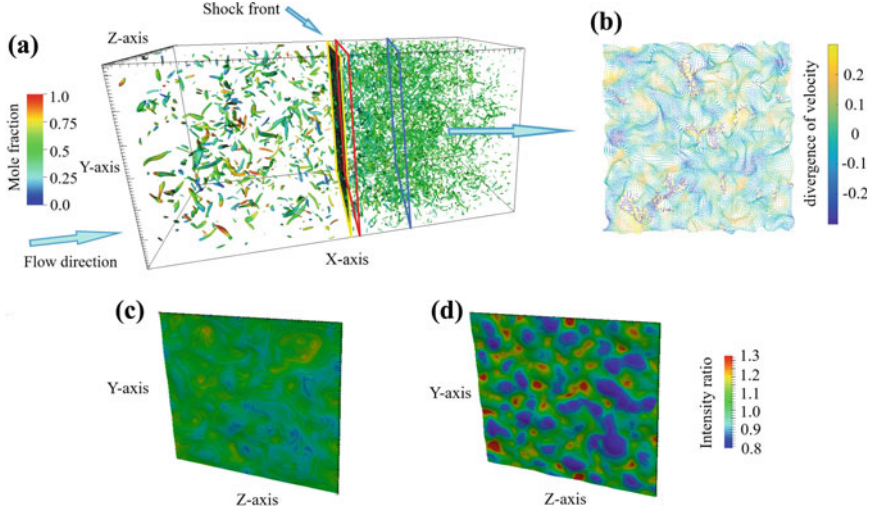


Fig. 1 Instantaneous contours of vorticity and shock surface in isotropic turbulence interacting with a shock wave and the shock Mach number is 2.0. **a** Vortex structure, identified by the Q criteria (i.e. isosurface of the second invariant of VGT: $Q = 2 \langle Q_w \rangle$, where $\langle Q_w \rangle$ is the averaged magnitude of rotation tensor) and colored by the mole fraction of the heavy fluid. Fluid particles are initialized as a sheet that spans over the homogeneous directions at a given post-shock streamwise position and allowed to develop with the flow. **b** Visualized particle sheet, convected and distorted by the post-shock turbulence. The instantaneous shock surface is colored by the shock intensity across the shock for **c** single-fluid and **d** multi-fluid cases

2.2 Lagrangian Method

For the current study, we have tracked more than 4.5 million particles that are initialized uniformly at various streamwise positions $\vec{x}_0$, and calculated various turbulence structure statistics following their trajectories. The aim is to understand the evolution of flow structures following fluid particles. Figure 1a marks with red lines a typical streamwise plane where particles are initialized. The particles are then convected by the instantaneous turbulent velocity obtained by turbulence-resolving shock-capturing simulations [15] and moved to another plane marked by the blue lines. At this stage, the initially flat particle sheet is distorted by the turbulence as shown in Fig. 1b.

The fluid particles are non-inertial and follow the local flow velocity. The corresponding transport equations for particle positions x_i^+ are:

$$\frac{dx_i^+(t|\vec{x}_0, t_0)}{dt} = u_i^+(t|\vec{x}_0, t_0), \quad (1)$$

$$u_i^+(t|\vec{x}_0, t_0) = u_i(x_i^+, t), \quad (2)$$

where $x_i^+(t|\vec{x}_0, t_0)$ represents the positions of the particles at time t that are initialized at $\vec{x}_0$ and time t_0 . The particle velocity $u_i^+(t|\vec{x}_0, t_0)$ can be obtained from the Eulerian velocity field $u_i(x_i^+, t)$ by interpolation. The interpolation is based on the cubic spline scheme, whose accuracy in predicting particle positions has been studied in Ref. [33]. In the STI configuration, there is a sharp change of the flow velocity at the shock, which deteriorates the interpolation accuracy. To achieve accurate interpolation of the particle velocity, the domain is partitioned into three different regions as shown in Fig. 1a: pre-shock, shock, and post-shock regions. The instantaneous shock surface is identified using the sensor: $s = -\theta/(|\theta| + \langle \omega_i \omega_i \rangle_{yz}^{0.5}) > 0.5$, where $\theta = \partial u_i / \partial x_i$ is the dilatation, $\omega_i = \epsilon_{ijk} \partial u_k / \partial x_j$ is the vorticity, and $\langle \rangle_{yz}$ represents the instantaneous average over the homogeneous directions. After the instantaneous shock region is identified, the pre- and post-shock turbulence fields can be separated for interpolation. Lagrangian dynamics of particles across the shock wave is not considered in this study.

3 Numerical Accuracy

The accuracy of the numerical results is addressed in this section through a series of convergence tests. To ensure that all the turbulence length scales are well resolved, a grid convergence test was conducted in Ref. [15]. Here, we summarize these results for completeness, together with additional convergence results for small-scale quantities. Figure 2 shows the velocity dissipation rate ε and mass fraction dissipation rate ε_ϕ as a function of the normalized streamwise direction $k_0 x$ for a series of meshes. The grey regions in the following figures indicate the unsteady shock region, inside which the results are affected by the shock wrinkling and unsteady shock movement. As the grid is refined in all three directions, both quantities display convergence,

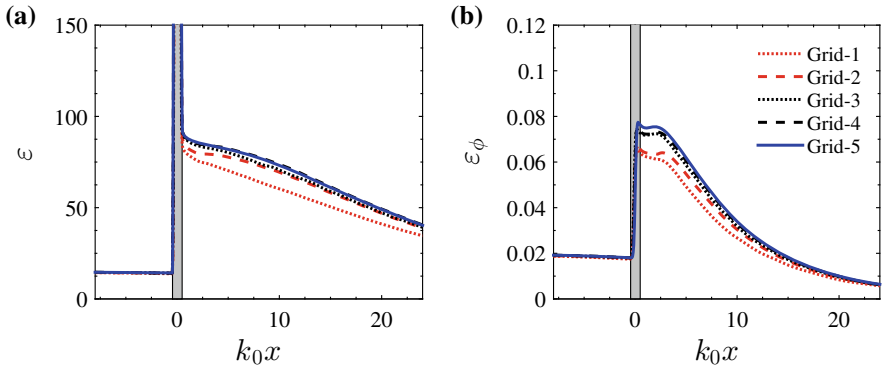


Fig. 2 Results of multi-fluid grid convergence tests at $Re_\lambda = 42$ and $M_t = 0.09$. Streamwise development of **a** velocity dissipation rate ε and **b** mass fraction dissipation rate ε_ϕ is shown. The region of unsteady shock movement is marked in grey. Grid numbers for Grid 1–5 are $256 \times 256 \times 1024$, $384 \times 384 \times 1024$, $384 \times 384 \times 1536$, $512 \times 512 \times 1536$, $512 \times 512 \times 2048$

proving the accuracy of the turbulence database. Another aspect that needs to be considered is the scale separation between the numerical shock thickness δ_n and the Kolmogorov length scale η as suggested in our previous study [15]. With the finest mesh ($512 \times 512 \times 2048$), the scale separation ratio (η/δ_n) is around 1.9, which is sufficient for resolving the interaction between the numerical shock wave and small-scale turbulent motions. Therefore, in the current study, we have obtained all the statistics from the turbulence field based on the finest grid to ensure accuracy. Secondly, LIA convergence tests were conducted following Ref. [11] to show that the shock-capturing simulations can capture the correct limits. Turbulent Mach number (M_t) and Taylor Reynolds number (Re_λ) were varied for the canonical single-fluid simulations, covering a wide range of parameter space. The shock-capturing simulation results do converge to LIA predictions for individual Reynolds stress components as long as certain conditions are satisfied [15]. This was the first time that the asymptotic values for individual Reynolds stresses were approximated using shock-capturing simulations.

Statistical convergence is another important factor that needs to be addressed before any further analysis. To reduce the statistical variability, all the results that are based on the Eulerian data are space-averaged over homogeneous directions and time-averaged for around two pass-over times.

For the Lagrangian statistics, the number of fluid particles needs to be large enough for statistical convergence, especially for conditional averaged statistics. The conditional statistics are obtained by ensemble averaging over all the fluid particles that fall into the bins (see Eq. (14)). The number of samples needed to achieve statistical convergence will be examined for different bin sizes. Figure 3 shows the convergence of two important conditional Lagrangian statistics $\langle \frac{DQ}{Dt} \rangle / \langle Q_w \rangle^{3/2}$, $\langle \frac{DR}{Dt} \rangle / \langle Q_w \rangle^2$ and their standard deviation (see Sect. 4.3 for definitions), depending on the number of particles in each bin. For the multi-fluid case, we note that the convergence of both conditional means and standard deviations can be achieved when using around

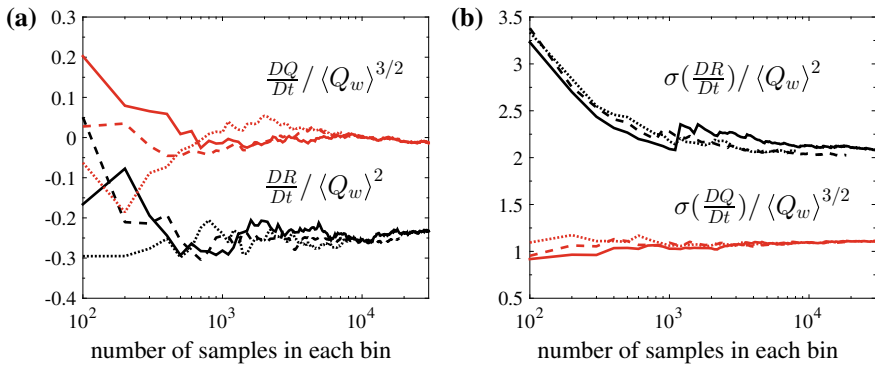


Fig. 3 The statistical convergence for **a** $\langle \frac{DQ}{Dt} \rangle / \langle Q_w \rangle^{3/2}$ and $\langle \frac{DR}{Dt} \rangle / \langle Q_w \rangle^2$ and **b** their standard deviations conditioned at point (3.0, 3.0) in the (Q, R) phase plane for multi-fluid case. The number of bins are 30×30 (solid), 40×40 (dashed) and 60×60 (dotted)

10,000 particles. The effects of the bin sizes are also examined by comparing three different set of bin numbers 30×30 , 40×40 and 60×60 in the (Q, R) phase plane at the same point $(3.0, 3.0)$. These bin numbers correspond to the following bin sizes: $(1.3, 1.3)$, $(1.0, 1.0)$ and $(0.67, 0.67)$. Our analysis indicate that the statistics converge to almost the same value when the sample size is large enough. In the present study, we uniformly sampled more than 4.5 million particles and made sure that there are at least 10,000 particles in each sample bin with the number of bins being 40×40 ($(\Delta Q, \Delta R) = (1.0, 1.0)$).

4 Results and Discussions

In this section, the effects of density variations on STI are examined in detail by comparing the results obtained from variable density (VD) or “multi-fluid” cases with those obtained from a reference ‘single-fluid’ simulation. The single-fluid reference simulation was conducted using the same inflow conditions for turbulent variables except density. In this reference case, the mass fraction of the heavy fluid is set to 1.0. At the same time, a passive scalar equation, which is the same as the mass fraction equation in the multi-fluid case, is solved for comparison. This case is referred to as just the single-fluid case and used as a reference to study the effects of VD on STI. For all cases, the turbulence is allowed to adjust itself to the scalar field in the pre-shock region before interacting with the normal shock. Statistical averages are computed over homogeneous directions to obtain statistics of the flow. Reynolds averages are denoted by an overbar, $\overline{f}$, while Favre averages are denoted by a tilde, $\tilde{f}$; the corresponding fluctuations around these averages are denoted by f' and f'' .

4.1 General Effects of Density on STI

Averaged flow statistics are compared in Fig. 4 for single-fluid and multi-fluid cases. Before the shock wave, all cases yield the same results. This observation not only confirms that the inflow conditions are somewhat similar in these cases, but also implies that for current simulation, the effect of density variations on turbulence is small in the pre-shock region. When comparing the multi-fluid turbulent kinetic energy and vorticity variance with the corresponding single-fluid values in Fig. 4a, d, it is noted that the amplification in these turbulent statistics is much more significant in the multi-fluid cases. Furthermore, the multi-fluid turbulent kinetic energy reaches a peak around $k_0 x \approx 2.0$, which is closer to the shock than $k_0 x \approx \pi$ for single-fluid case. Figure 4b, c show the comparison for streamwise turbulence Taylor micro length scale, λ_1 , and Kolmogorov length scale, η . The reduction in turbulence length scales across the shock wave is evident in these figures; the multi-fluid cases show more reduction than the single-fluid case. Note that the changes in turbulence statistics

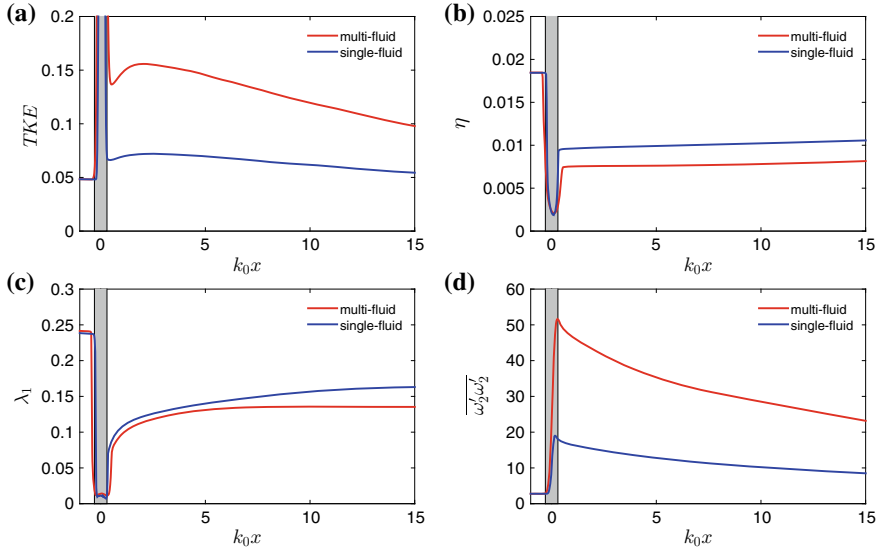


Fig. 4 Plots of **a** turbulent kinetic energy, **b** Kolmogorov length scale, **c** Taylor micro scale and **d** transverse vorticity variance for multi-fluid (red) and single-fluid (blue) simulations. Figure reprinted with permission from Tian et al. [15]

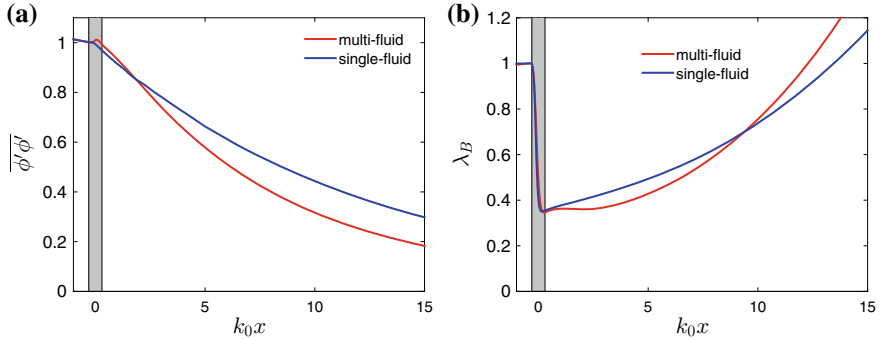


Fig. 5 Plots of normalized **a** scalar variance, **b** Batchelor scale for multi-fluid (red) and single-fluid (blue) simulations. Figure reprinted with permission from Tian et al. [15]

in multi-fluid cases are expected to depend on the scalar structure and the Atwood number [34]; these are not discussed in this paper.

In Fig. 5, statistics related to the scalar field (heavy fluid mole fraction for multi-fluid case and passive scalar for the single-fluid case) and mixing are compared. Both scalar variance $\overline{\phi'\phi'}$ and Batchelor scale λ_B are shown. λ_B is calculated based on the scalar dissipation and is the representative of the smallest scales in the scalar field. The scalar statistics are normalized by the values immediately before the shock wave. After passing through the shock wave, the faster decay of scalar variance for the multi-

fluid case indicates stronger shock enhancement of scalar mixing. The Batchelor scale, however, shows a more complex behavior. Unlike the Kolmogorov length scale, the same reduction ratio of Batchelor scale across the shock wave is observed between the multi-fluid and single-fluid cases. After passing through the shock wave, the Batchelor scales of multi-fluid cases exhibit a transient process of decreasing before returning to the pre-shock value, during which an even smaller structure of the scalar field is generated. In the single-fluid case, however, the Batchelor scale increases monotonically back to its pre-shock value. We also note that after $k_0 x \approx 10.0$, the multi-fluid λ_B values are larger than the single-fluid values as the faster mixing immediately after the shock smooths out the small scalar scales.

To further understand the variable density effects, conditional expectations of several turbulence quantities, conditioned on the density, are calculated and examined. In Fig. 6a, the conditional expectation of the turbulent kinetic energy, TKE, is shown. We note that TKE has a preferential distribution in the relatively high or low density regions in the post-shock regions. One possible explanation is that in high and low density regions, the local sound speed has different values from that of average sound speed, so that the local shock velocity, $u_{1,s}$, becomes nonzero (in the reference frame of the laminar shock wave) and changes significantly. The local movement of the shock surface then further changes the post-shock velocity, $u_{1,d}$ and makes it deviate from the averaged post-shock velocity $\overline{u_{1,d}}$. The deviation from $\overline{u_{1,d}}$ or u'_i in low and high density regions is much larger in magnitude than that in regions with moderate density, which results in larger TKE in the high and low density regions. We have computed the conditional average of u'_i on density. Results agree very well with our explanation for conditional TKE. We also note that TKE is larger in the light fluid regions compared to heavy fluid regions. This is due to the low inertia of the light fluid, which responds faster to the changes in the local strain and, thus, accelerates faster [35, 36]. This explanation is also applicable to Fig. 6b for single-fluid case, which shows a preferential distribution of TKE in the lower density fluid region before the shock wave. After passing through the shock

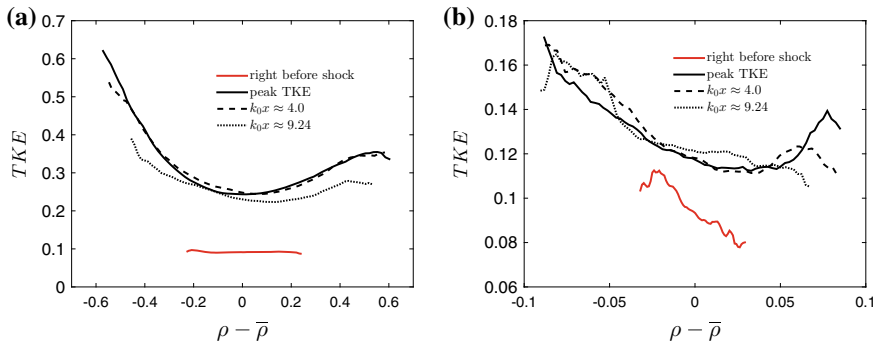


Fig. 6 Conditional expectation of TKE as a function of density at various streamwise location for: **a** multi-fluid and **b** single-fluid case. Figure reprinted with permission from Tian et al. [15]

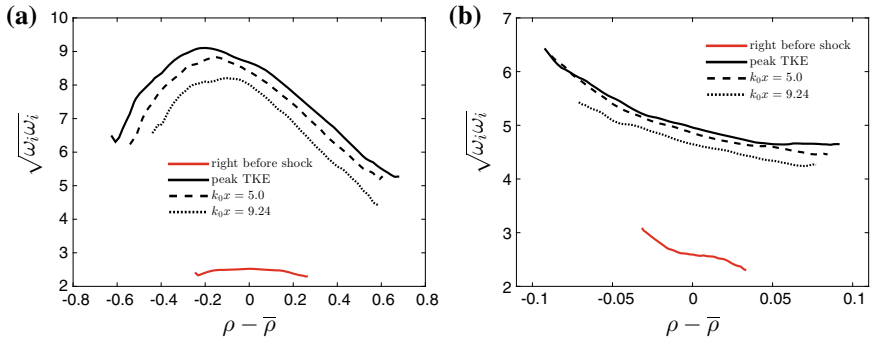


Fig. 7 Conditional expectation of enstrophy as a function of density at various streamwise location for: **a** multi-fluid and **b** single-fluid case. Figure reprinted with permission from Tian et al. [15]

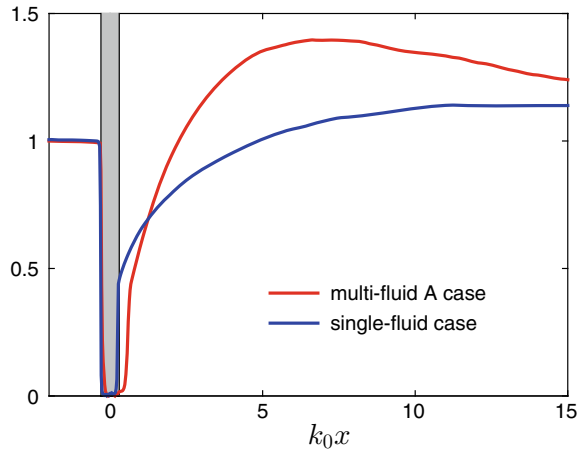
wave, a stronger amplification in the high and low density regions is noted but this is relatively weak.

The correlation between vorticity and density has a completely different behavior than TKE. Evidently, Fig. 7 shows that more vorticity is generated in the mixed fluid regions. Before reaching the shock wave, the mixing process is relatively slow, so that there are still large regions with pure or partially mixed fluids and only narrow regions with fully mixed fluids. In these regions, the density gradients remain large. Through the interaction with the shock wave, the large density gradients lead to the generation of vorticity through the baroclinic torque. For the single-fluid case (Fig. 7b), the distribution of vorticity is not affected much by the shock because of the absence of large density variations.

4.2 Structure and Topology of the Post-shock Turbulence

References [1, 3, 9, 12] showed that STI leads to a two-dimensionalization of the flow immediately after the shock, i.e. the flow becomes locally axis-symmetric. The vortex stretching term is therefore the smallest immediately after the shock. This term then increases as the flow evolves away from the shock and returns to a 3-D structure. Following the expression of the vortex-stretching term in Ref. [12], the turbulent stretching term can be decomposed into contributions from eigenvectors and eigenvalues of strain rate. After normalizing the turbulent stretching term in the enstrophy equation using $\overline{\omega_2' \omega_2'}$ and the turbulence time scale TKE/ϵ (where ϵ is the rate of dissipation), the effects of the eigenvectors and eigenvalues of the strain rate can be isolated. As shown in Fig. 8, the magnitude of turbulent stretching decreases to around zero across the shock wave. During the transient post-shock process, the multi-fluid case exhibits a faster return to 3-D isotropic turbulence structure, indicated by the faster increase in the normalized turbulent stretching term. Further

Fig. 8 Vortex-stretching term in the enstrophy equation, normalized by $\overline{\omega_2 \omega_2}$ and turbulence time scale TKE/ϵ . Figure reprinted with permission from Tian et al. [15]



downstream, the multi-fluid case reaches its peak value much sooner than the single-fluid case. The behavior of the normalized turbulent stretching term indicates that the contributions to the return to 3-D turbulence not only come from increased enstrophy amplification, but also from the change of alignment between vorticity and strain rate tensor eigenvectors.

The preferential amplification of the transverse components of the rotation and strain rate tensors is a well-known effect in STI and has been extensively studied for the canonical single-fluid flows [7, 11, 12]. This amplification can lead to an increase in the correlation between the two quantities. When density variations are introduced, Ref. [15] showed that the transverse components of vorticity variance are further enhanced across the shock wave while the streamwise component retains almost the same value. To better understand the variable density effects on post-shock turbulence, the PDF of the strain-rotation angle, Ψ , is considered in Fig. 9. Ψ is calculated using $\Psi = \tan^{-1}(S_{ij}S_{ij}/(W_{ij}W_{ij}))$, where $S_{ij} = 1/2(A_{ij} + A_{ji})$ and $W_{ij} = 1/2(A_{ij} - A_{ji})$ are the strain and rotation tensors and $A_{ij} = \partial u_i / \partial x_j$ is the velocity gradient tensor. In isotropic turbulence, the PDF of Ψ peaks at 90° [37], indicating a strain dominated flow. In single-fluid post-shock turbulence, the PDF of Ψ exhibits a shift of the peak from 90° to around 45° , as the shock Mach number increases. This has been observed by Livescu and Ryu [12] and is interpreted as the increase in correlation of strain and rotation. However, in the multi-fluid case, the peak still occurs at relatively large angles and the increase in correlation is not as pronounced as that in the single-fluid case, at the same shock Mach number. Figure 9 implies that the rotation and strain are amplified differently by the shock when large density variations are present, which compromises the correlation between the above two quantities.

To understand the effects of the density variations on the strain and rotation tensors as the flow interacts with the shock wave, the pre- and post-shock values of the conditional expectations of the magnitudes of the two tensors are shown in Fig. 10 for

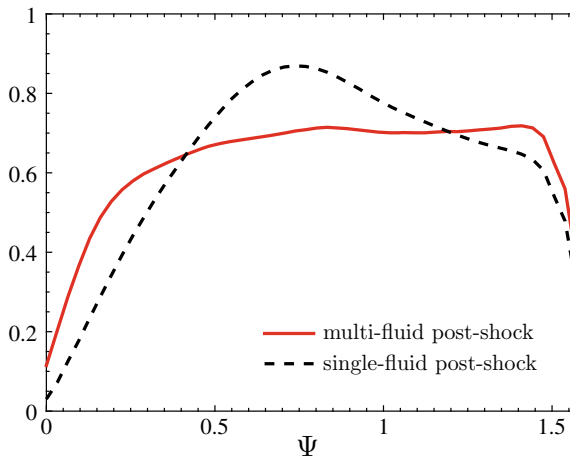


Fig. 9 PDF of the strain-entropy angle Ψ in degrees for post-shock turbulence

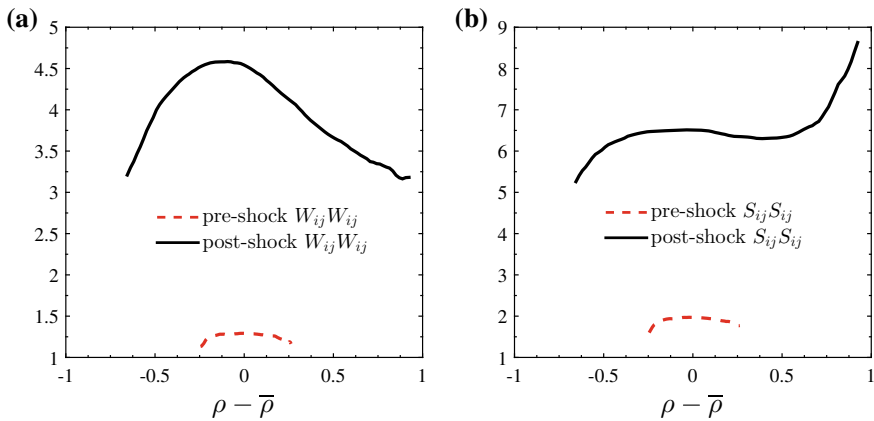
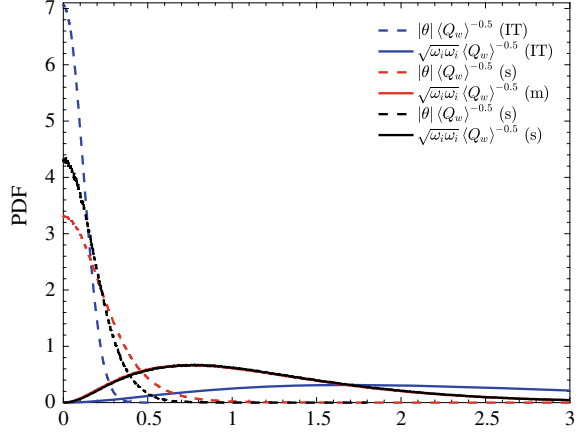


Fig. 10 Conditional expectation of the magnitude of **a** rotation tensor and **b** strain rate tensor as a function of density before and after the shock

the multi-fluid case. The amplification of vorticity is stronger in the mixed fluid region as shown and explained in the last section. On the other hand, Fig. 10b shows that the magnitude of the strain rate tensor tends to be stronger in the heavy fluid regions and weaker in the light fluid region. This trend is hypothesized to be related to the dependence of shock strength on the pre-shock density. Reference [15] showed that shock strength is positively correlated with density. In the highest density regions, the shock compression is also stronger, while it is weaker in the smallest density regions, leading to the observed trend in the amplification of the strain rate tensor magnitude. This trend is different from that observed for the vorticity, which is explained above. As a result, the trend of the strain-entropy angle PDF peaking around 45° , observed

Fig. 11 PDFs of the normalized dilatation and vorticity for isotropic turbulence (IT), single-fluid post-shock turbulence, and multi-fluid post-shock turbulence



in the single fluid case at higher shock Mach numbers is weakened in the multi-fluid case.

To further characterize the turbulence structure behind the shock wave, we have analyzed the invariant space of the VGT. The second and third invariants (denoted by Q^* and R^*) of the anisotropic/deviatoric part of the VGT can reveal important features of the flow topology [38]. In highly compressible turbulence, there exists a richer set of flow topologies due to the dilatational part of the velocity gradient tensor [39]. For the parameter range considered in this study; however, the compressibility effects are weak. This is demonstrated in Fig. 11, where the normalized PDFs of the dilatation and vorticity for pre-shock turbulence, single-fluid, and multi-fluid post-shock turbulence are shown. The pre-shock isotropic turbulence has a very low magnitude of dilatation. The shock wave expectedly amplifies the dilatation magnitude, and more so when variable density effects exist, but the dilatation values are still considerably lower than those studied in Refs. [39–41]. Considering that the focus of this study is on the variable density effects, here we only present the topological structure of the anisotropic velocity gradient tensor, using data points where $P \approx 0$. These regions encompass about 60% of the flow. The anisotropic part of the VGT is calculated using the formula $A_{ij}^* = A_{ij} - \theta/3I$. Correspondingly, the second and third invariants can be calculated from:

$$Q^* = -\frac{1}{2} A_{ij}^* A_{ji}^*, \quad (3)$$

$$R^* = -\frac{1}{3} A_{ij}^* A_{jk}^* A_{ki}^*, \quad (4)$$

Based on the local values of Q^* and R^* , four types of local flow topologies can be identified: stable-focus/stretching (SFS), unstable-focus/contracting (UFC), stable-node/saddle/saddle (SN/S/S) and unstable-node/saddle/saddle (UN/S/S). For isotropic turbulence, the joint PDF of (Q^*, R^*) has the well-known tear-drop shape.

This has been further observed in other fully developed turbulent flows, such as boundary layers, mixing layers, and channel flows [38, 42]. This type of distribution of Q^* and R^* is an indicator that the turbulence is more likely having a local topology of stable-focus/stretching or an unstable-node/saddle/saddle. In Fig. 12a, it is shown that the joint PDF of normalized second and third invariants, $Q^*/\langle Q_w \rangle$ and $R^*/\langle Q_w \rangle^{3/2}$, has the same tear-drop shape in the pre-shock flow. Using shock-LIA and DNS data, Ref. [11] showed that for single-fluid STI, the (Q^*, R^*) distribution is significantly modified by the shock wave, with a tendency towards symmetrization of the joint PDF. This indicates that the regions with stable-focus/compression and stable-node/saddle/saddle (first and third quadrant) are more likely to occur as turbulence develops a 2-D axisymmetric flow structure. To understand the variable density effects on this shock-induced symmetrization, the joint PDFs of (Q^*, R^*) for both single-fluid and multi-fluid post-shock turbulence are compared in Fig. 12b, c.

Figure 12b shows the joint distribution for the single-fluid post-shock turbulence. The dashed lines denote the locus of zero discriminant of A^* , where Q^* and R^* satisfy $27R^{*2}/4 + Q^{*3} = 0$. Compared to the pre-shock joint PDF, there is a tendency towards symmetrization, with more points located in the first and third quadrants. This agrees very well with previous shock-LIA results [11]. Similar to single-fluid STI, multi-fluid STI demonstrates a tendency towards symmetrization of the (Q^*, R^*) distribution. However, the multi-fluid distribution is slightly more symmetric and has a larger variance, with more points away from the axes. This implies that more extreme “events” exist in the post-shock multi-fluid turbulence. It also agrees with previous results on multi-fluid STI that shows amplification of TKE to be stronger when there are significant density variations [15, 16].

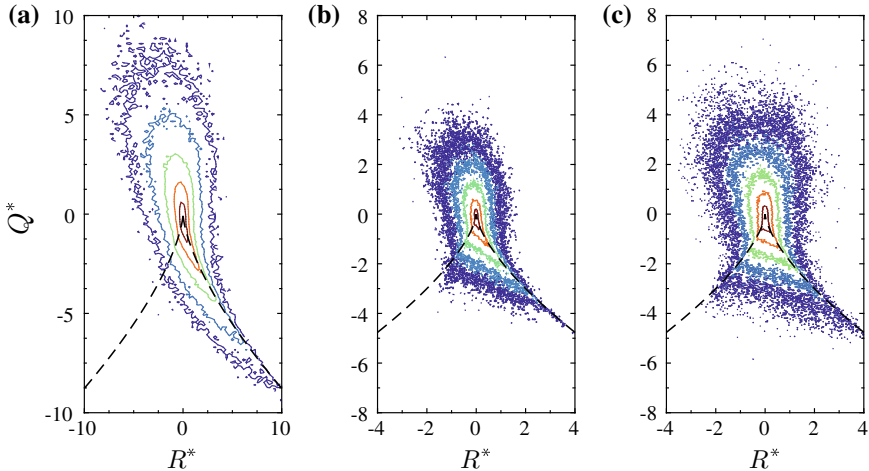


Fig. 12 Iso-contour lines of joint PDFs of normalized second and third invariants of the anisotropic part of the velocity gradient tensor, (Q^*, R^*) , for **a** pre-shock, **b** single-fluid post-shock turbulence, and **c** multi-fluid post-shock turbulence. The lateral lines denote the locus of zero discriminant

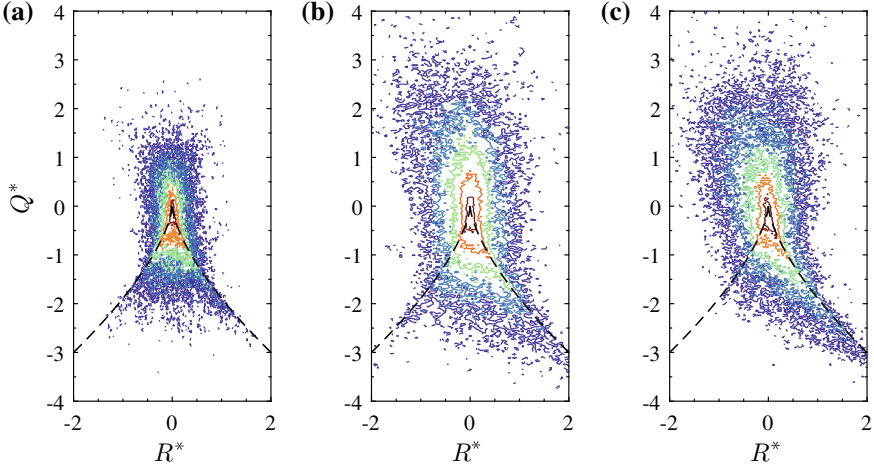


Fig. 13 Iso-contour lines of post-shock ($k_0x \approx 0.44$) joint PDF of second and third invariants of the anisotropic part of the velocity gradient tensor, (Q^*, R^*) , in regions with different densities. **a** Regions with high density values, $\rho > (\bar{\rho} + 90\% \rho'_{rms})$, **b** regions with density around the post-shock mean value, and **c** regions with low density values, $\rho < (\bar{\rho} - 90\% \rho'_{rms})$

The density effects on the post-shock joint PDF of second and third invariants are further explored by comparing the conditional distribution, conditioned on regions with different densities, in Fig. 13a–c. Figure 13a corresponds to regions with relatively high density ($\rho > (\bar{\rho} + 90\% \rho'_{rms})$), Fig. 13b to regions with density around the post-shock mean value, and Fig. 13c to low density regions ($\rho < (\bar{\rho} - 90\% \rho'_{rms})$). For consistency check, the joint PDFs corresponding to these regions are also computed for the pre-shock flow (not shown) and found to be close to the single-fluid PDFs. After the shock wave, the joint PDFs demonstrate significant differences between regions with different densities. In regions with density closer to that of post-shock mean density, the distribution of invariants appears to be very similar to that shown in Fig. 12c. But for regions with higher density (Fig. 13a), the joint PDF becomes more symmetric compared to the overall flow or single-fluid case. There is a much larger portion of data points having a local topology of stable-node/saddle/saddle, and fewer data points fall into the first and second quadrants, indicating larger strain-dominated regions. On the other hand, the post-shock regions with low-density values (Fig. 13c) exhibit features similar to that of isotropic turbulence, with almost the same tear-drop shape, only with a larger variance or a wider distribution. The quantitative difference is hypothesized to be related to the higher shock strength variation in the multi-fluid case. It was observed in our previous studies [43], that the local shock strength is positively correlated with the pre-shock density. With a stronger shock, the two-dimensionalization effect on the post-shock turbulence should also appear stronger in the high-density regions [12]. For low-density regions, the smaller two-dimensionalization effect reduces the symmetrization trend. Moreover, the relatively lower inertia in these regions leads to a faster response to

the local strain field [36], which could make a faster return to isotropic turbulence. The different characteristics of (Q^*, R^*) joint PDF in regions with different densities provide additional evidence for the previous argument made about the density role on the preferential amplification of the strain and rotation tensors.

4.3 Lagrangian Dynamics of VGT

Lagrangian equations of the VGT and its invariants can be used to understand the evolution of turbulence structure. The time evolution of A_{ij} for fluid particles can be obtained by taking the spatial derivatives of the Navier-Stokes equations. In dimensionless form, it can be written as [40]:

$$\frac{DA_{ij}}{Dt} = \frac{\partial A_{ij}}{\partial t} + u_k \frac{\partial A_{ij}}{\partial x_k} = -A_{ik}A_{kj} - H_{ij} + \mathcal{T}_{ij}, \quad (5)$$

with

$$H_{ij} = \frac{\partial}{\partial x_j} \left(\frac{1}{\rho} \frac{\partial p}{\partial x_i} \right) = -\frac{1}{\rho^2} \frac{\partial \rho}{\partial x_j} \frac{\partial p}{\partial x_i} + \frac{1}{\rho} \frac{\partial p^2}{\partial x_i \partial x_j} = H_{ij}^b + H_{ij}^p, \quad (6)$$

$$\mathcal{T}_{ij} = \frac{\partial}{\partial x_j} \left(\frac{1}{\rho} \frac{\partial \sigma_{ik}}{\partial x_k} \right), \quad \sigma_{ij} = \frac{\mu}{Re_0} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right), \quad (7)$$

where Re_0 is the reference Reynolds number. From here, the dynamic equations for the three invariants of the VGT, P , Q , and R can be derived in the following form [40]:

$$\frac{DP}{Dt} = (P^2 - 2Q) + H_{ii}^p + H_{ii}^b - \mathcal{T}_{ii}, \quad (8)$$

$$\begin{aligned} \frac{DQ}{Dt} = & (PQ - 3R) + (PH_{ii}^p + A_{ij}H_{ji}^p) + (PH_{ii}^b + A_{ij}H_{ji}^b) \\ & + (-P\mathcal{T}_{ii} - A_{ij}\mathcal{T}_{ji}), \end{aligned} \quad (9)$$

$$\begin{aligned} \frac{DR}{Dt} = & PR + (QH_{ii}^p + PA_{ij}H_{ji}^p + A_{ij}A_{jk}H_{ki}^p) + (QH_{ii}^b + PA_{ij}H_{ji}^b \\ & + A_{ij}A_{jk}H_{ki}^b) + (-Q\mathcal{T}_{ii} - PA_{ij}\mathcal{T}_{ji} - A_{ij}A_{jk}\mathcal{T}_{ki}), \end{aligned} \quad (10)$$

where the three invariants of VGT are defined as:

$$P = -tr(A_{ij}), \quad (11)$$

$$Q = \frac{1}{2}(tr(A_{ij})^2 - tr(A_{ij}A_{jk})), \quad (12)$$

$$R = -det(A_{ij}), \quad (13)$$

Here, $tr(A_{ij})$ and $det(A_{ij})$ denote the trace and determinant of a tensor. Note that instead of the deviatoric part of the VGT, the dynamic equations for the full VGT are considered. The reason is that due to the variable density effects and shock compression, the incompressibility condition is not satisfied especially when M_t and A_t become large. Even though M_t and A_t in this study are small, we still consider the full equations for any future comparisons. The contributions from the dilatational part of the VGT and their coupling with the variable density effects in highly compressible turbulence are still unknown and need to be explored for STI in future studies.

The dynamical equations can be divided into contributions by four different parts: (I) mutual-interaction among invariants, (II) pressure Hessian, H_{ij}^p , (III) baroclinic, H_{ij}^b and (IV) viscous term $\mathcal{T}_{ij}$. The statistics regarding these terms can be extracted from the Lagrangian data.

The Lagrangian dynamics of the turbulence and the evolution of flow topology are examined by considering the conditional mean rate of change of Q and R in the invariants plane [22]. The rates of change are used to form a vector at each point in the invariants plane. The trajectories implied by these vectors can be followed to understand the return-to-isotropy. In fully compressible turbulence, the (P, Q, R) invariant space becomes three-dimensional [39–41] and there exists an out-of-plane (Q, R) component of the trajectory due to the contribution from compressibility (P) effect. Due to the low compressibility effect in this work, however, it would be more appropriate to consider only the in-plane (Q, R) dynamics and leave the compressibility effects for future study. Therefore, the results presented below correspond to the data points with small magnitude of P ($|P|/\langle Q_w \rangle^{0.5} < 0.1$) for the relatively “incompressible” region of the flow. These points comprise approximately 60% of the flow. The procedure used to obtain the conditional mean vectors (CMVs) in this study is similar to that in Ref. [22]. The condition averages are computed using the formula:

$$\begin{aligned} \langle X | (A = A_0, B = B_0) \rangle &= \langle X | (A_0 - \frac{1}{2} \Delta A) \leq A < (A_0 + \frac{1}{2} \Delta A), \\ &\quad (B_0 - \frac{1}{2} \Delta B) \leq B < (B_0 + \frac{1}{2} \Delta B) \rangle, \end{aligned} \quad (14)$$

therefore, $X(Q, R)$ represents a statistical quantity that is conditioned on Q and R .

The normalized conditional mean vectors $(DQ/Dt/\langle Q_w \rangle^{3/2}, DR/Dt/\langle Q_w \rangle^2)$ for different flows are shown in Fig. 14. The vectors obtained from isotropic turbulence data are shown in Fig. 14a for reference. It can be seen that the CMVs exhibit a circulating behavior in the (Q, R) plot around the origin in the clockwise direction, indicating that the flow evolves from SFS to UFC, UN/S/S, SN/S/S then back to SFS on average. This circulating behavior represents the Lagrangian dynamics in fully developed turbulence that maintains the tear-drop shape of the (Q, R) distribution. This has been observed in many incompressible/compressible canonical turbulent flows [22, 40]. The CMVs for single-fluid and multi-fluid post-shock turbulence are shown in Fig. 14b, c. Evidently, the joint PDF of (Q, R) becomes more symmetric due to shock compression. From the Lagrangian point of view, the circulating

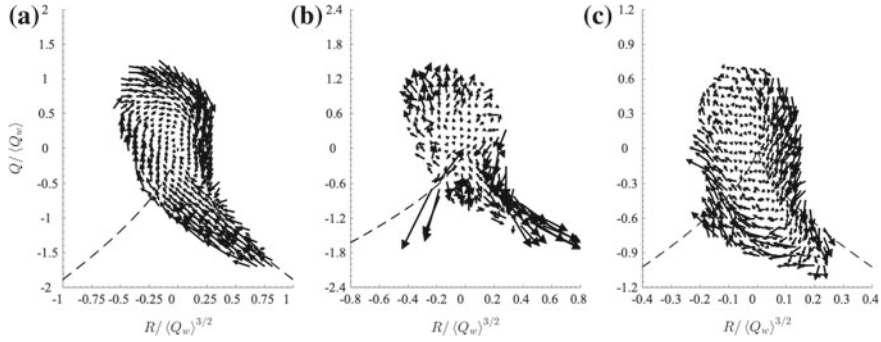


Fig. 14 Conditional mean rate of change vectors of $(DQ/Dt / \langle Q_w \rangle^{3/2}, DR/Dt / \langle Q_w \rangle^2)$ in the (Q, R) plane for **a** isotropic turbulence, **b** single-fluid post-shock turbulence, and **c** multi-fluid post-shock turbulence at streamwise location of $k_0 x \approx 0.5$. To ensure that the vectors can be properly visualized, their sizes are re-scaled by multiplying with a constant of 0.3. This applies to all the following vector plots

behavior as seen in Fig. 14a isotropic turbulence is weakened. The particles in Q_2 tend to have an increasing Q and decreasing R , resulting in an overall trend of getting away from the original point, instead of circulating and then moving toward Q_1 . This trend in the second quadrant represents an increase of enstrophy. The particles in Q_1 have similar dynamics as in isotropic turbulence and tend to move downward in the (Q, R) plane toward the zero discriminant curve. The particles in Q_3 are more likely to move straight up towards Q_2 , while those in Q_4 are likely to move away from the original point following the direction of the zero discriminant line and then circulate back to Q_3 . The overall behavior formed by these particles demonstrates the return-to-isotropy process, with an enlarging head in the second quadrant and elongating tail in the fourth quadrant, anticipating the formation of the classic tear-drop shape.

The density effects can be further examined by conditioning the $(DQ/Dt, DR/Dt)$ vector field on the local density. Figure 15a shows the CMVs for the light fluid regions. The light fluid particles retain the circulating motion, except that the particles in Q_3 and Q_4 are likely to go straight left instead of following the zero discriminant line. In general, the flow dynamics in the light fluid regions are less affected by the shock wave. For the medium density fluid regions (Fig. 15b), the circulating motion disappears. On the right side of the (Q, R) plane ($R > 0$), which is the strong dissipation-production region, the fluid particles tend to move downward, resulting in lower Q values. On the left side of the (Q, R) plane ($R < 0$), which is the enstrophy-production dominated region, the fluid particles tend to move to the left, indicating an increased enstrophy-production. The overall downward-moving behavior of the medium density fluid particles is indicative of decreasing vorticity. This is possibly due to the fact that vorticity is preferentially amplified in the medium density region across the shock wave. After passing the shock wave, the vorticity will decrease as the correlation between density and vorticity vanishes. Figure 15c shows the CMVs for the heavy fluid regions. Interestingly, the heavy fluid particles exhibit

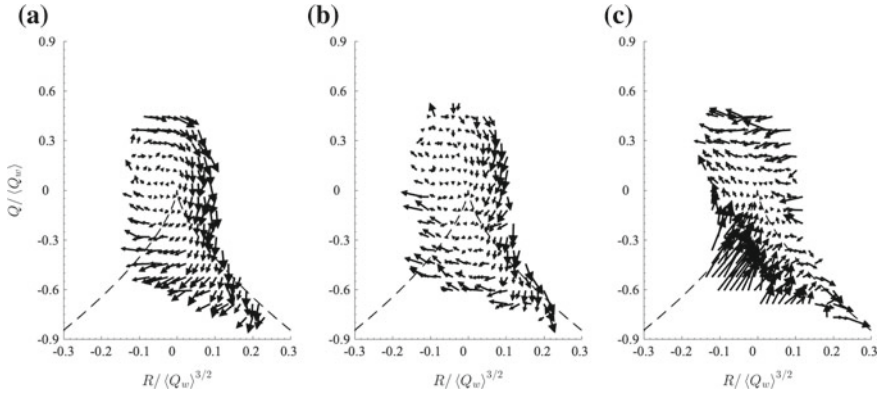


Fig. 15 Conditional mean vectors in the (Q, R) invariants plane for **a** light fluid, **b** medium density fluid and **c** heavy fluid at streamwise location of $k_0 x \approx 0.5$

counterclockwise motion. The heavy particles start from Q_3 and move to Q_4 , Q_1 , and finally to Q_2 . This implies that they become vorticity dominated due to the fast depletion of strain. Evidently, density plays an important role in the development of the flow topology in the post-shock region, so special attention should be made to the modeling of variable density STI.

To better understand the underlying mechanisms that cause the behavior highlighted above, the dynamic Eqs. (9) and (10) governing the vector $(DQ/Dt, DR/Dt)$ are examined. The Lagrangian dynamics of the flow can be understood better by considering the conditional mean vectors of different terms in the (Q, R) plane. As a reference, these terms are shown in Fig. 16 for isotropic turbulence. The variable Q tends to be amplified in the enstrophy-production dominated region due to the effects of vortex stretching mechanism and is decreased in the dissipation-production dominated region due to self-amplification of the strain rate tensor. On the other hand, the mutual effects on R are small because the first invariant P is usually small and the positive and negative values are likely to cancel each other. The contributions from the pressure Hessian (Fig. 16b) tend to move the particles away from an asymptotic line, ending up amplifying the magnitude of R . This result agrees well with that observed in turbulent boundary layers [40]. For the current simulation, the asymptotic line starts from Q_2 and ends in Q_4 with a slope of around -2.5 . The baroclinic contributions are very small in the post-shock turbulence as shown in Fig. 16c. The viscous effects as shown in Fig. 16d and as expected are reducing the magnitudes of Q and R and pushing the particles towards the origin. This has been observed in various types of turbulence [22, 40]. The combined effects from the four above mechanisms determine the circulating behavior of the conditional mean of $(DQ/Dt, DR/Dt)$ vectors.

For multi-fluid post-shock turbulence, the pressure Hessian term is also the only term that is qualitatively different than that in isotropic turbulence (Fig. 17). Despite the increased density and pressure gradient in the multi-fluid case, the baroclinic term

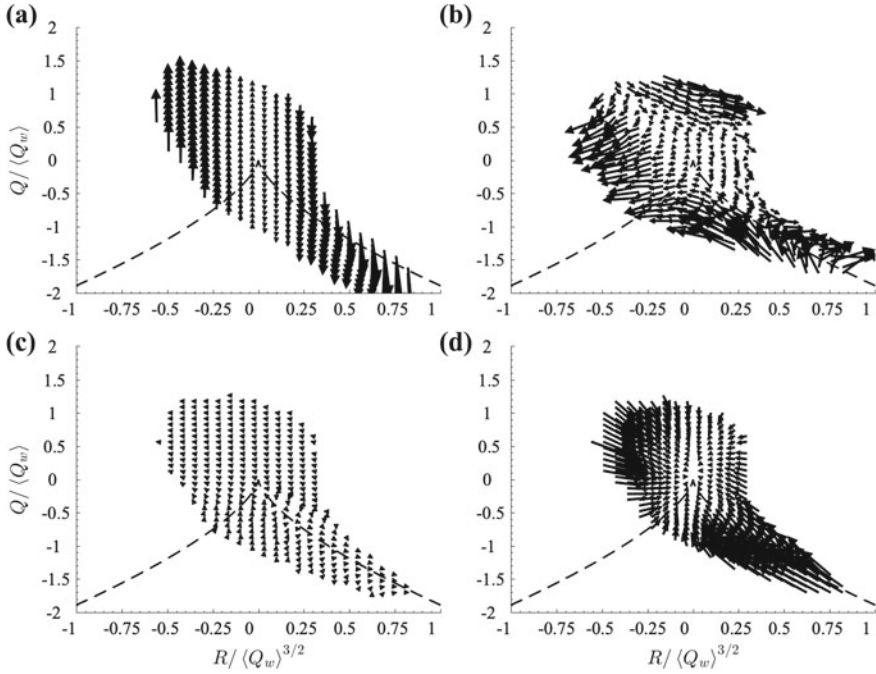


Fig. 16 Contributions to the transport equations of the VGT invariants by different terms for isotropic turbulence. **a** Mutual interaction among invariants, **b** pressure Hessian term, **c** baroclinic term, and **d** viscous term

is still considerably smaller than all the other terms. In Q_2 and Q_3 , an asymptotic line similar to that in isotropic turbulence seems to exist, which “repels” the vectors away from it, causing an increase in $|R|$ values. In Q_1 and Q_4 , the magnitude of pressure hessian term becomes much smaller. The further conditioned pressure Hessian term based on the local densities in Fig. 18 indicates that fluid particles with different densities have very different behaviors with respect to pressure Hessian dynamics. Specifically, the pressure Hessian generally moves the heavy particles toward the regions with larger Q values. In Q_3 and Q_4 , it also moves the heavy fluid particles towards the $R > 0$ plane. For the light fluid particles, the pressure Hessian term tends to make them move towards regions with larger $|R|$ values in the first and second quadrant. In Q_3 and Q_4 , the fluid particles move from Q_4 to Q_3 . Last but not the least, the fluid particles with medium density seem to exhibit similar behavior to light fluid particles, except in Q_1 , where the pressure Hessian contribution is moving the fluid particles towards the regions with large Q values. Examining Figs. 15 and 18 together, we observe that the differences in particle dynamics in the (Q, R) plane in regions with different densities are mainly due to differences in the pressure Hessian contributions.

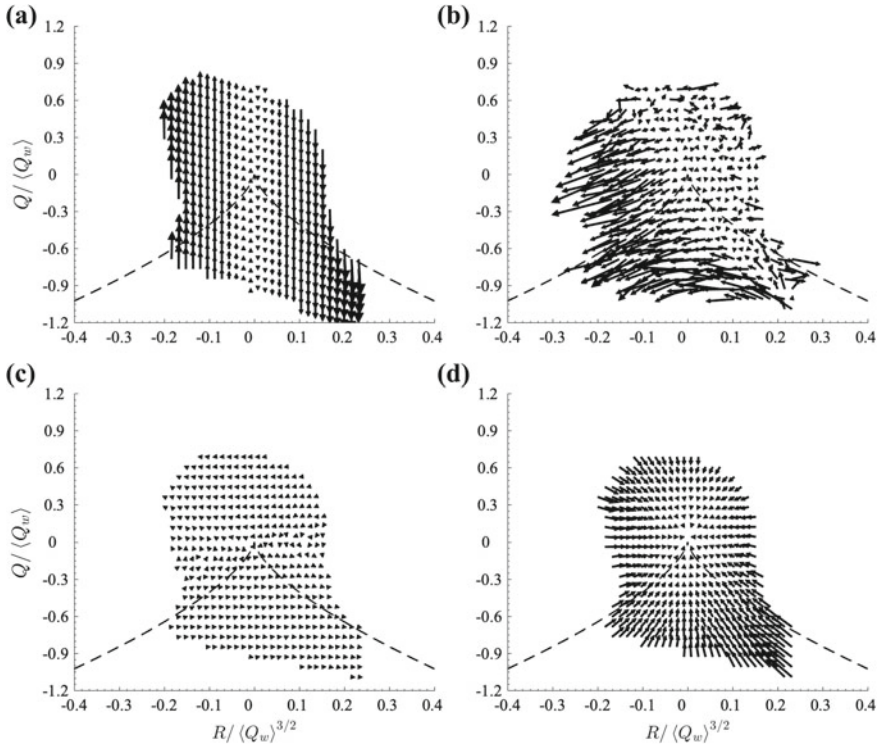


Fig. 17 Contributions to the dynamics of the VGT invariants by different terms for multi-fluid post-shock turbulence. **a** Mutual interaction among invariants, **b** pressure Hessian term, **c** baroclinic term, and **d** viscous term

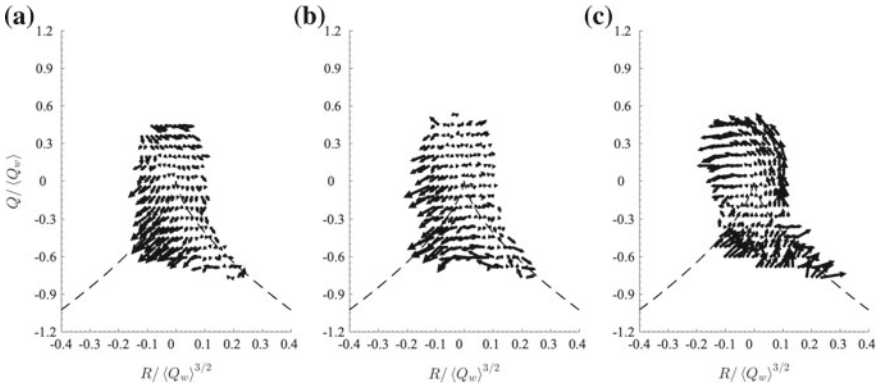


Fig. 18 Contributions from pressure Hessian to the dynamics of the VGT invariants in **a** light fluid region, **b** medium density fluid region and **c** heavy fluid region

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